GEOG 172

Think Geographically, Measure Locally

Lecture 7-1

LOCAL STATISTICS

These tests are designed to test the null hypothesis of randomness around a prespecified point – typically against an alternative that there is a clustering of raised risk around the point.

- Calculated from subsets of the data 'in the neighborhood' of each location
- Global statistics are made from the pieces of local statistics
- Often used in exploratory analyses – finding patterns and outlier regions

Geometry and Topology

Area Relationships Revisited

NATURE OF THE SPATIAL SUBSET

Boots & Okabe (2007) identify two different kinds of local spaces and then characterize the relationships that exist between local areas $\{A_1, \ldots, A_n\}$ within a study region R using two important properties.

Focused Local Space

Local space that is determined ***with respect to a given location***. For example, a circular area drawn around a point location.

Unfocused Local Space

Local space that is determined ***with respect to an attribute*** rather than location. For example, a subset of zones defined by population density.

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Exhaustive

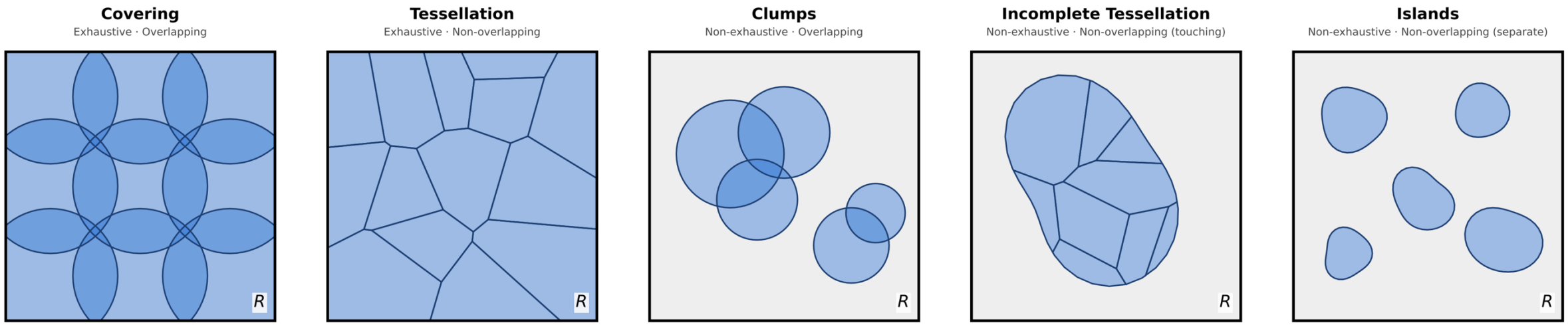
Local spaces cover the entire study area

Overlapping

Local spaces overlap with respect to one another

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Grey square = study region R; blue shapes = areas A_i (darker blue where areas overlap; grey where R is not covered).

GETIS-ORD G

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Gi Statistic

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Captain Getis' Buried Treasure

If only there were a stat for that

ACTIVITY

Overview: A legendary pirate, Captain Getis, has hidden his fortune somewhere on SpatStat Island. Before disappearing, he left behind a mysterious treasure map showing traces of gold scattered across the island. Can you find his treasure?

Scenario:

- SpatStat Island is a 20x30 grid where each cell contains a number of gold coins. However, only some of the treasure is observable from the surface.
- Several treasure-hunting crews have arrive on the island, but only have time to dig in one location.
- To find the treasure, crews must work through three clues, identify promising locations, and use local Getis-Ord statistics to decide where to dig. **The crew with the closest statistically justified excavation wins.**

The Spatstat Island

Captain Getis' Sunken Treasure







	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
B	0	0	0	0	1	0	0	0	1	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
D	1	0	1	3	0	2	0	1	0	0	1	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
E	0	0	1	2	2	0	0	0	1	0	0	1	0	3	0	1	0	1	0	0	0	0	0	0	3	0	0	0	0	0
F	0	0	0	0	0	0	1	0	0	0	1	1	1	0	0	1	1	2	1	0	0	1	0	3	5	2	0	0	0	1
G	0	0	0	0	0	0	0	1	1	1	1	0	2	2	2	3	5	0	0	2	0	3	1	2	5	2	0	0	0	0
H	0	0	0	0	0	0	2	0	1	2	6	3	3	3	5	4	7	6	7	1	1	0	0	1	0	1	0	0	0	0
I	0	0	0	0	0	1	0	0	0	1	2	1	6	6	5	6	9	7	2	3	0	2	1	2	0	0	0	0	0	0
J	0	0	0	0	0	0	0	0	0	0	0	2	2	8	10	9	8	12	5	5	2	0	0	0	0	0	0	1	0	0
K	0	0	0	0	0	0	0	0	1	1	0	0	3	8	9	11	8	15	11	2	1	1	0	0	0	0	0	0	0	0
L	0	0	1	0	0	0	0	0	0	0	2	0	8	1	5	5	8	7	6	3	1	0	1	0	0	0	0	1	0	0
M	0	0	0	0	0	0	0	0	0	1	1	2	2	5	3	6	4	4	5	1	2	0	0	1	4	0	1	1	0	0
N	0	0	0	0	0	0	0	0	1	2	3	1	0	3	4	1	2	2	3	2	1	0	0	2	4	1	1	0	0	0
O	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	2	1	1	0	0	0	0	1	3	3	0	1	0	0	0
P	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0
Q	0	0	0	1	3	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
R	0	0	0	2	0	0	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
S	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
T	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0



All That Glitters is not Gold

"The greatest treasure leave more than a single trace. Seek the out where riches gather, for one glittering square may lead you to the wrong place"

- Look for areas where several nearby cells contain relatively high values, rather than just individual cells with the largest numbers.
- Circle the most promising gold-rich region on the map.
- Within the most promising region, select **three nearby candidate cells** that appear to be surrounded by relatively high gold readings. Record their coordinates (A, B, and C).

No Pirate Sales Alone

"A lone golden coin makes no fortune. Look beyond the sand beneath your feet, and count the riches in the surrounding portion."

- For each of the three candidate locations, record the number of coins in the focal cell.
- Add the gold readings in the **eight neighboring cells** to obtain the neighborhood sum.
- Calculate local G_i for each location using the total of 500 visible coins:
- Compare the three G_i values. Which candidate has the greatest concentration of gold in its surrounding neighborhood?

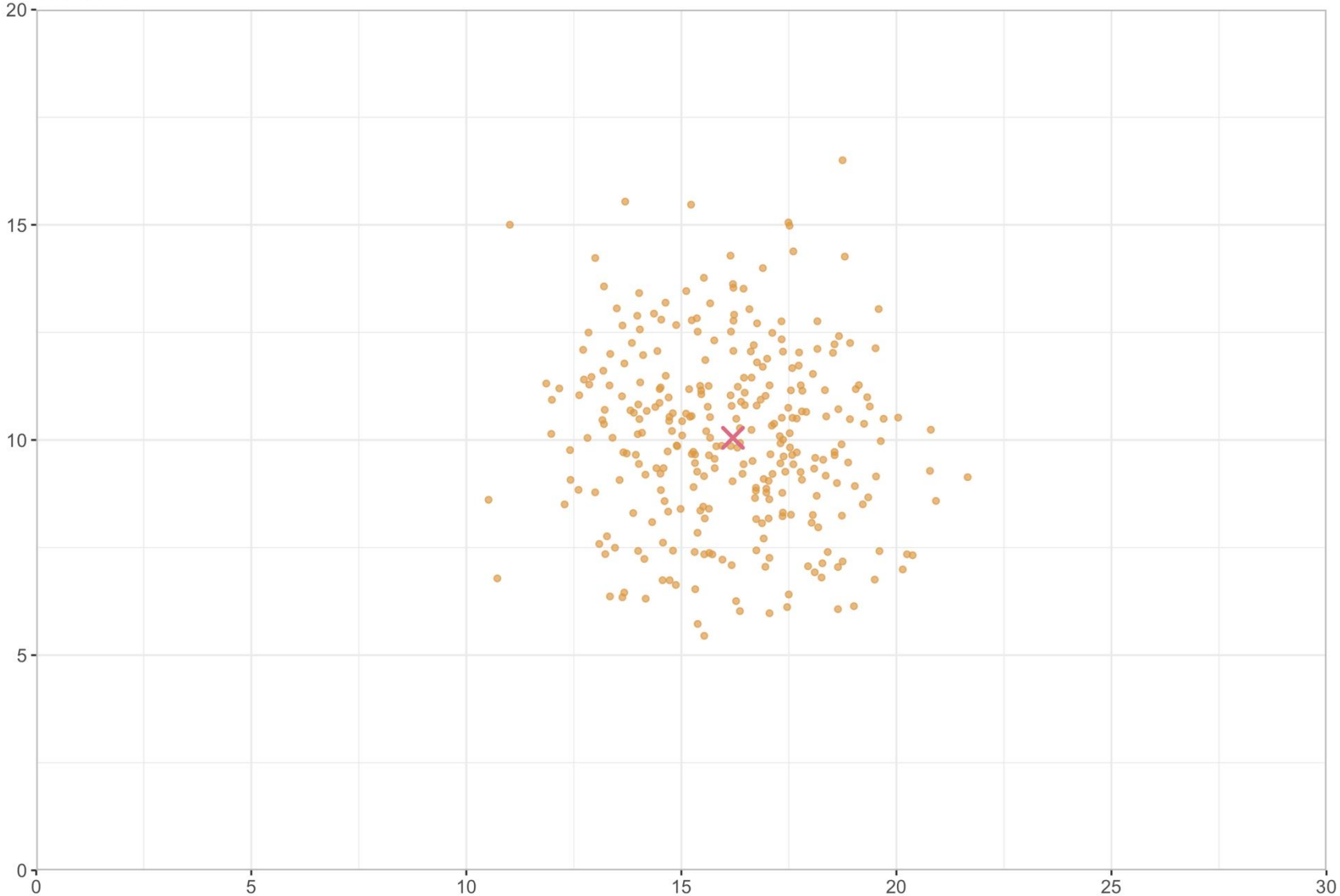
As the Geographer

"Many places contain gold, but a golden trail isn't always profound. Ask your geographer if what chance produced is unusual, and if true wealth has been found.

- Use the supplied values to calculate an approximate z-score for each location:
- Which candidate provides the strongest evidence of a local concentration of gold?
- Return to the map and select one final excavation site, using both the calculations and the surrounding readings to justify the decision. Crews may revise their initial choice, but should explain why.

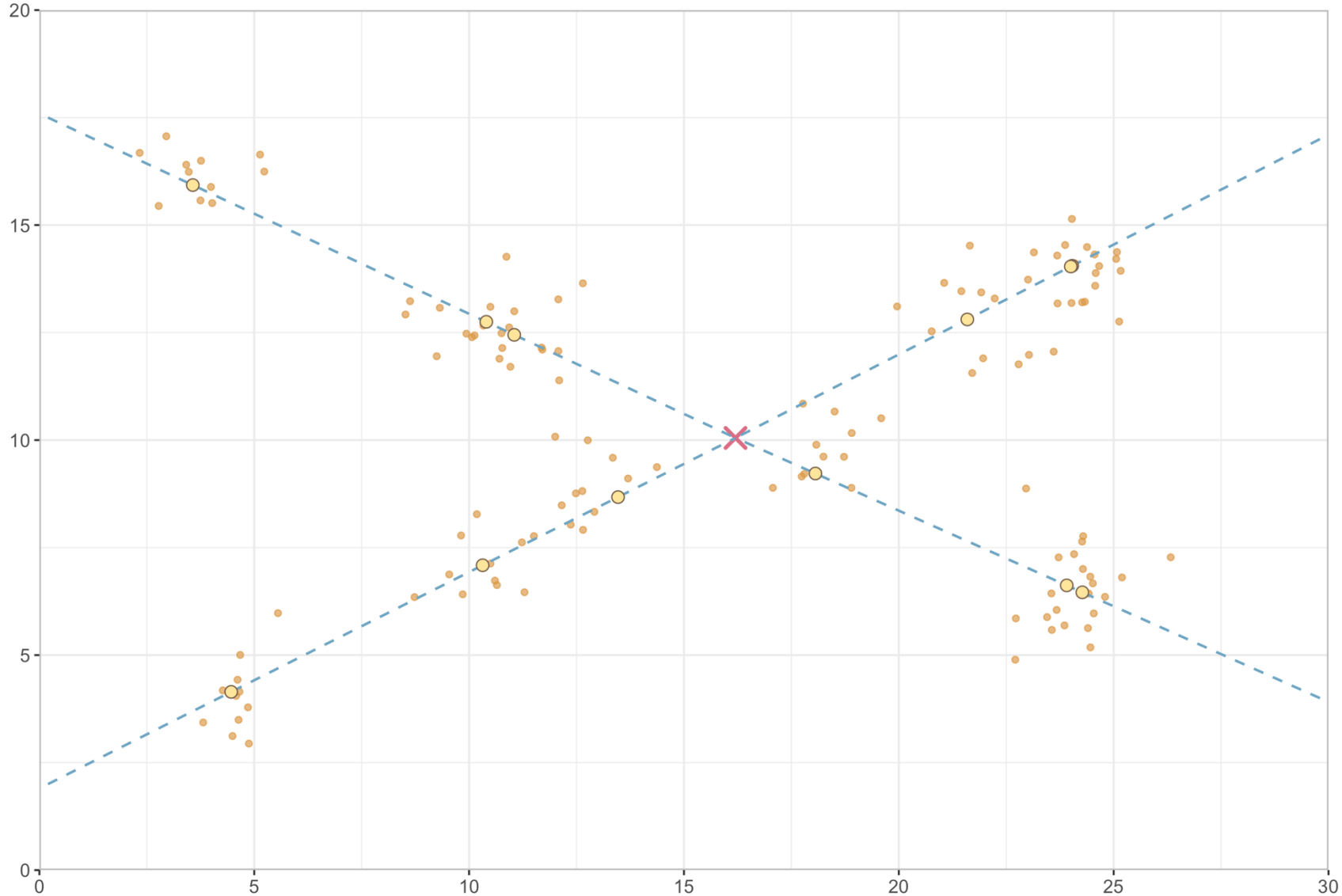
Process 1: Environmental Dispersal

300 coins



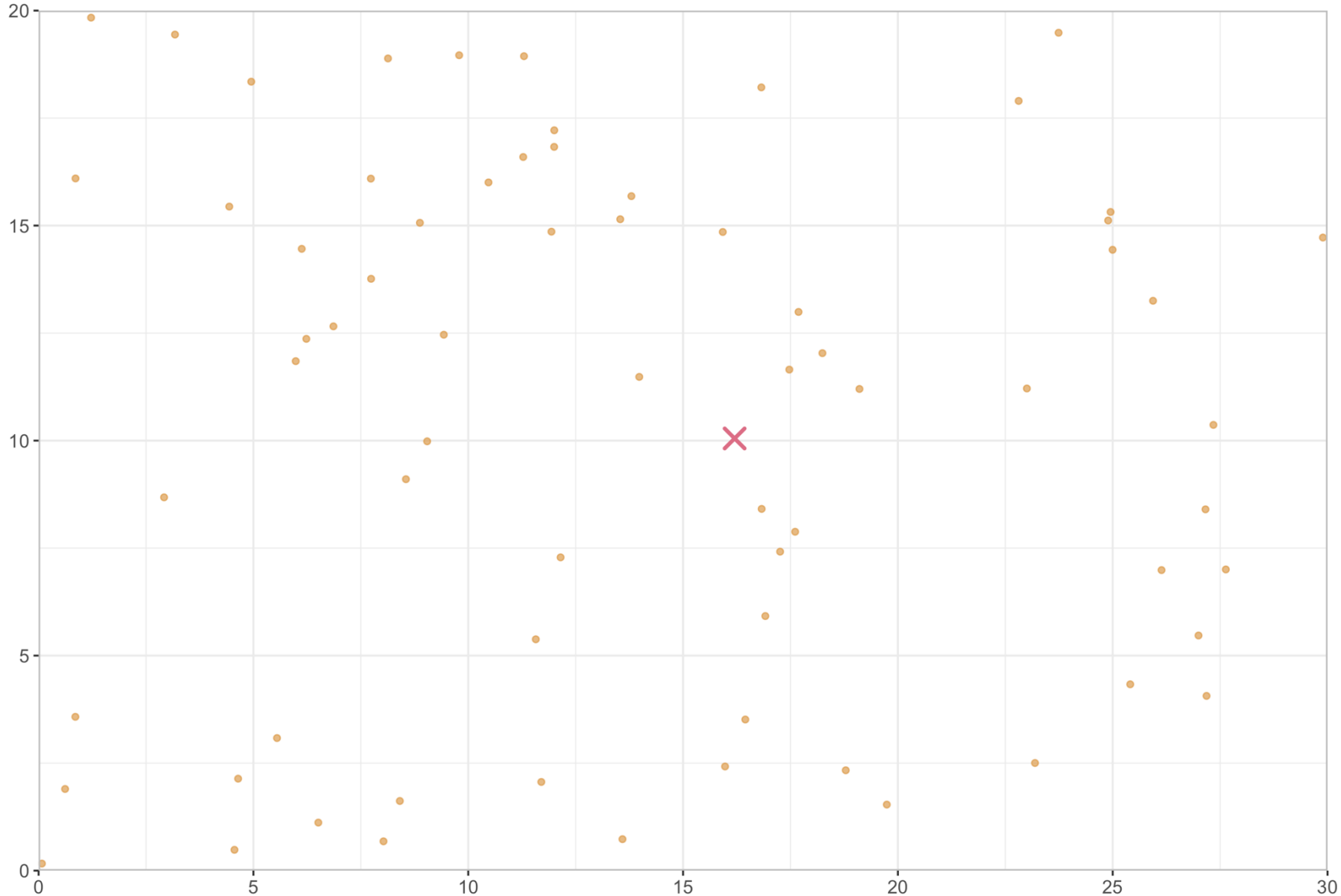
Process 2: Transportation Spills by Pirates

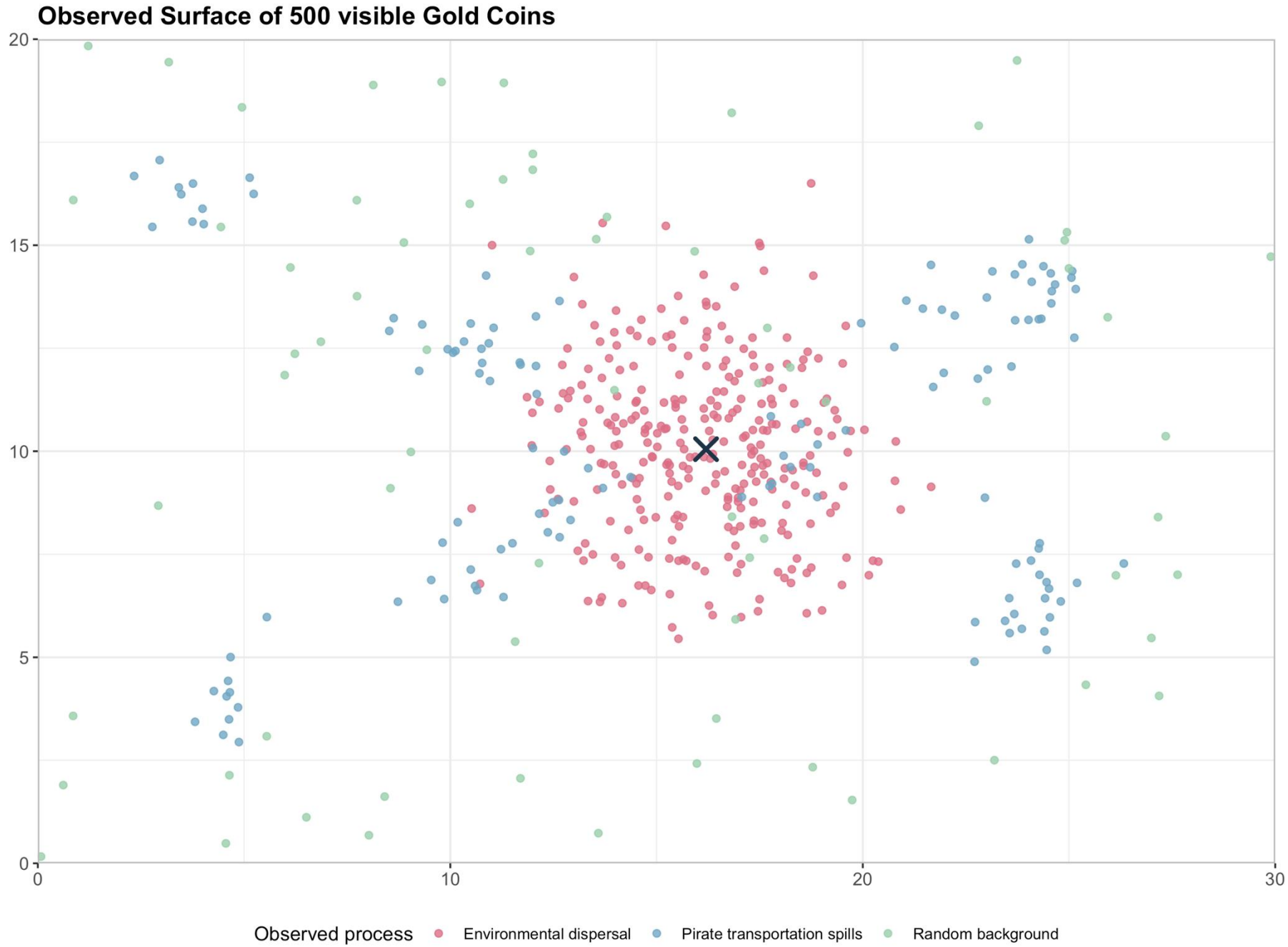
130 coins

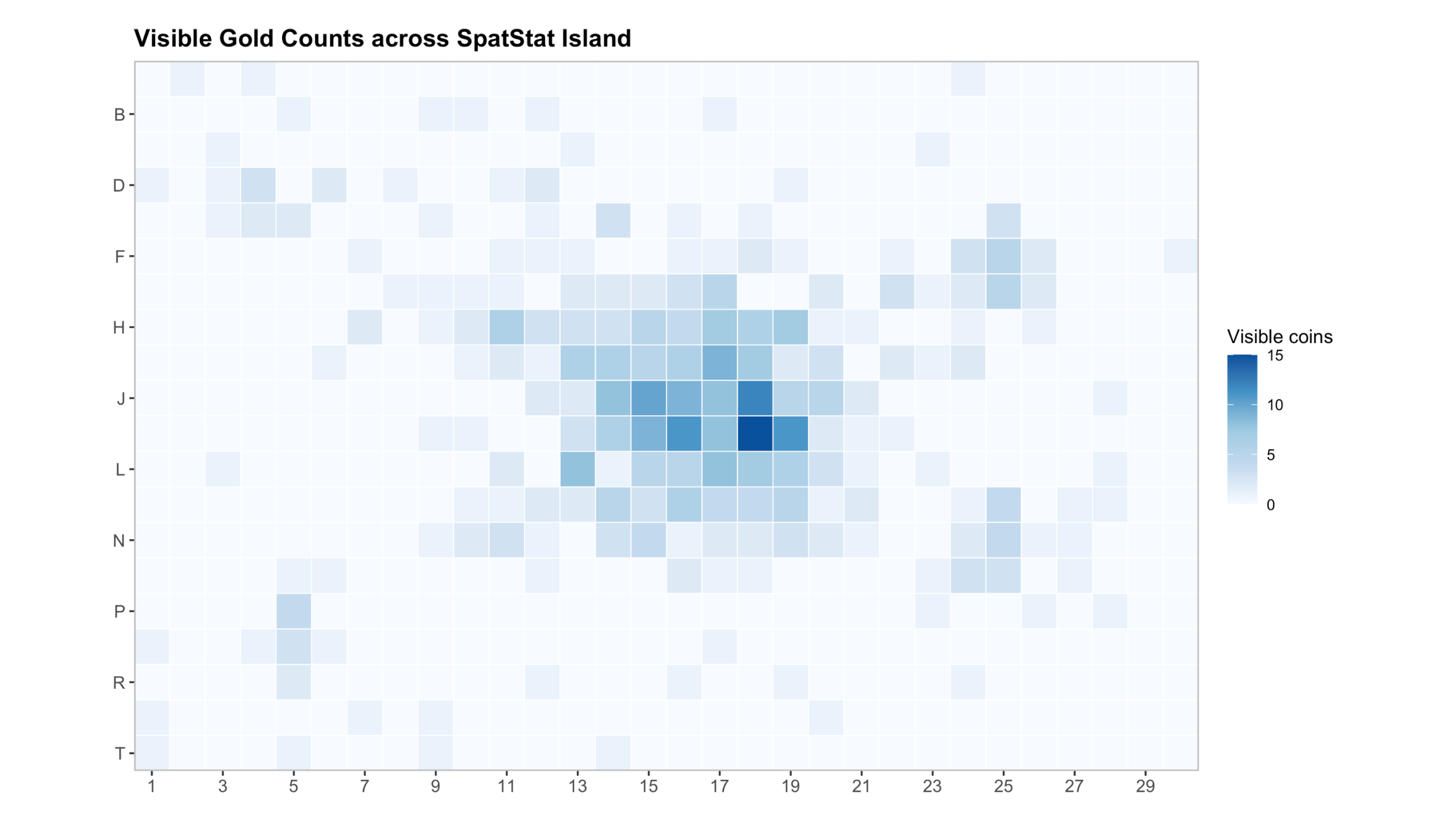


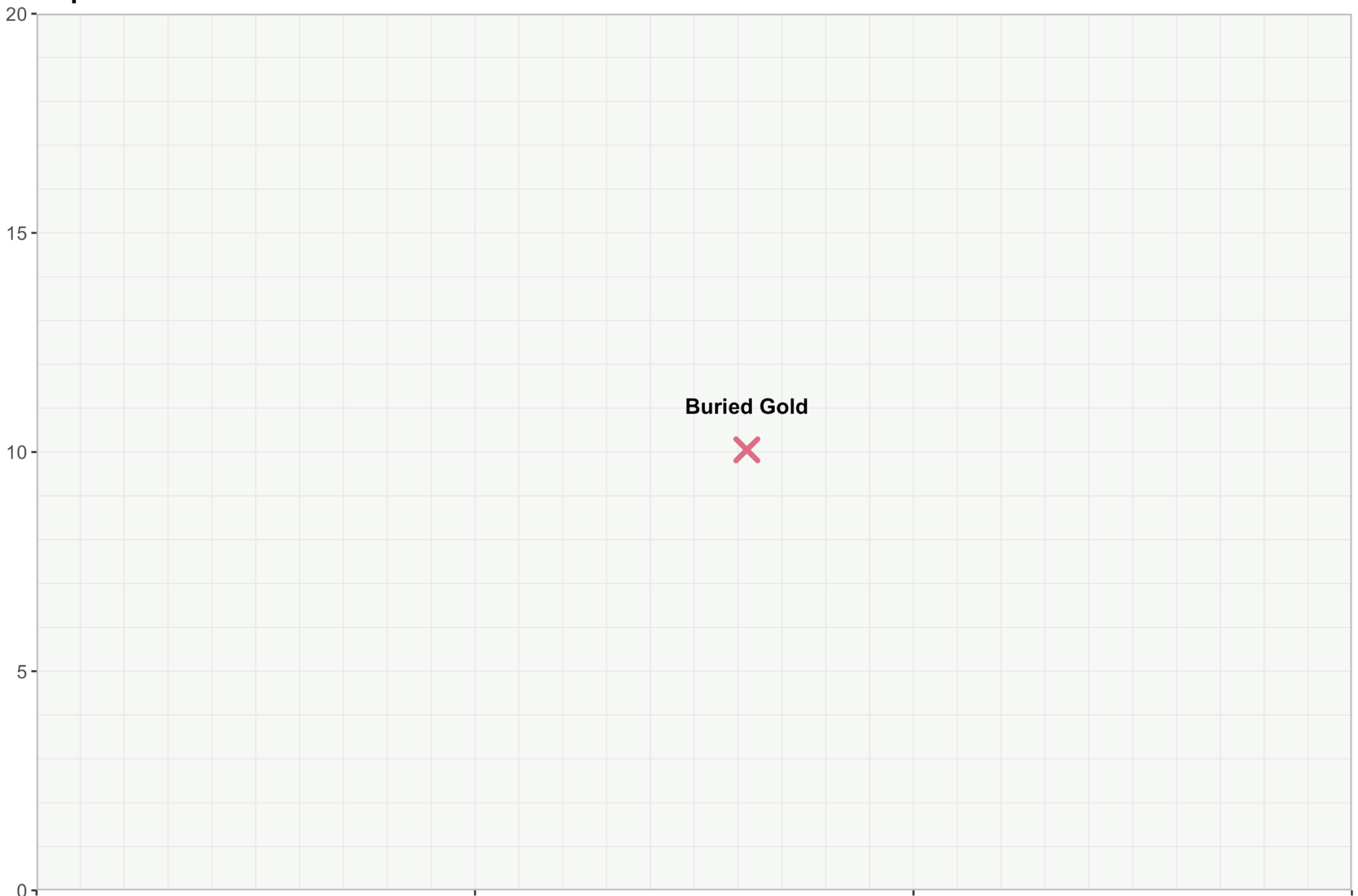
Process 3: Random Background

70 Coins




Observed Surface of 500 visible Gold Coins
20
15
10
5
0
0
10
20
30
Observed process
Environmental dispersal
Pirate transportation spills
Random background


Visible Gold Counts across SpatStat Island
B
D
F
H
J
L
N
P
R
T
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
Visible coins
15
10
5
0


20
15
10
5
0
Buried Gold

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IMPLEMENTATION IN R

```R
localG(x, listw, zero.policy = attr(listw, "zero.policy"), spChk = NULL,
       GeoDa = FALSE, alternative = "two.sided", return_internals = TRUE)
```

- x — A numeric vector of attribute values, one per location, in the same order as the neighbors list. This should be a ratio-scaled variable with a natural origin.
- listw — The spatial weights object (built with nb2listw()). This is where the analyst's definition of "local" (d) enters the calculation.
- alternative — The direction of the hypothesis test: "two.sided" (default, hot *or* cold spots), "greater" (hot spots only), or "less" (cold spots only).

IMPLEMENTATION IN R

```R
library(spdep)

nb <- poly2nb(tracts, queen = TRUE)                    # queen contiguity neighbors
lw <- nb2listw(nb, style = "B", zero.policy = TRUE)  # binary weights

tracts$Gi <- localG(tracts$income, lw)
```

```R
# Add each location to the neighbors with include.self for Gi-Star
lw_star <- nb2listw(include.self(nb), style = "B", zero.policy = TRUE)
tracts$Gi_star <- localG(tracts$income, lw_star)
```

TAKEAWAYS

- Local statistics break Global statistics into one value per location
- The geometry of the areas shapes what these statistics can tell us, and what corrections need to be made.
- All local statistics depend on how local is defined by the weights matrix